DIRECTEDNESS MAKES COLIMITS OF SETS PRESERVE SURJECTIONS ONTO PRODUCTS BUT NOT PRODUCTS

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Editorial note. Counted span is the author's own proof environment, source lines 2482-2510, following the statement at 2467-2480.

Editorial note. Part (2) asserts (G x G)/G is not isomorphic to a point; the orbit structure of the diagonal translation action (orbits isomorphic to G) is readable from the displayed data.

Allowed standard prerequisites (editorial).

- Colimits of set-valued diagrams as a quotient description. Section section-limit-sets of this chapter, referenced at proof lines 2487-2488. Elements of colimits represented by values at objects; the one-object category computes colimits as coinvariants.

Source reference. Original lemma 002W: sources/stacks-categories/pinned-original-source.tex, source lines 2399–2399. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original section 002U: sources/stacks-categories/pinned-original-source.tex, source lines 1807–1807. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{I}$ be an index category, i.e., a category. Assume that for every pair of objects x, y of $\mathcal{I}$ there exist an object z and morphisms $x \to z$ and $y \to z$. Then*

(1) *If M and N are diagrams of sets over $\mathcal{I}$, then* $\operatorname{colim}(M_i \times N_i) \to \operatorname{colim} M_i \times \operatorname{colim} N_i$ *is surjective,*
(2) *in general colimits of diagrams of sets over $\mathcal{I}$ do not commute with finite nonempty products.*

Proof. Proof of (1). Let $(\overline{m}, \overline{n})$ be an element of $\operatorname{colim} M_i \times \operatorname{colim} N_i$. Then we can find $m \in M_x$ and $n \in N_y$ for some $x, y \in \mathrm{Ob}(\mathcal{I})$ such that m maps to $\overline{m}$ and n maps to $\overline{n}$. See Section 002U. Choose $a : x \to z$ and $b : y \to z$ in $\mathcal{I}$. Then $(M(a)(m), N(b)(n))$ is an element of $(M \times N)_z$ whose image in $\operatorname{colim}(M_i \times N_i)$ maps to $(\overline{m}, \overline{n})$ as desired.

Proof of (2). Let G be a non-trivial group and let $\mathcal{I}$ be the one-object category with endomorphism monoid G. Then $\mathcal{I}$ trivially satisfies the condition stated in the

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lemma. Now let G act on itself by translation and view the G-set G as a set-valued $\mathcal{I}$-diagram. Then

$$\operatorname{colim}_{\mathcal{I}} G \times \operatorname{colim}_{\mathcal{I}} G \cong G/G \times G/G$$

is not isomorphic to

$$\operatorname{colim}_{\mathcal{I}}(G \times G) \cong (G \times G)/G$$

This example indicates that you cannot just drop the additional condition Lemma 002W even if you only care about finite products. □